Appendix II. Debt Sustainability Analysis Update

1. **This debt sustainability analysis (DSA) updates the analysis presented to the Board in September 2014 and indicates that Sierra Leone remains at moderate risk of debt distress**. Although domestic and external debt levels have risen since the last DSA update, all debt indicators are below their respective policy-dependent indicative thresholds throughout the projection period (2014–34). The analysis also indicates that the medium- to long-term debt outlook is vulnerable to adverse shocks to several macroeconomic variables, notably exports, commodity prices, foreign direct investment (FDI) inflows, government revenue, and borrowing conditions. Consequently, staff reiterates the need for continued prudent borrowing policies, sustained fiscal consolidation efforts, continued implementation of growth-enhancing structural reforms, and promotion of economic diversification. The authorities agreed with Staff's assessment and policy recommendations.

2. **The baseline macroeconomic assumptions underlying this DSA update are:**

   - **Economic growth is expected to slow from 20 percent in 2013 to 6.0 percent in 2014. It is projected to contract by 12.8 percent in 2015 due to the differential distribution of the impact of the Ebola epidemic and suspension of production in the iron ore sector with most of the aggregate effect in 2015**. Growth is expected to pick up in 2016 and average 7.3 percent for 2016–19. Over the long term, growth is expected to stabilize close to the historical rates for the non-iron ore economy of around 5.2 percent.

   - **Inflation is projected to rise in the medium term.** Disruption of economic activity due to the Ebola outbreak and depreciation of the Leone are expected to push up inflation from 8.3 percent in 2014 to 13.1 percent in 2015. Inflation is expected to decline gradually and hover around 5.4 percent in the long term.

   - **The overall fiscal deficit is projected to widen in the short term and then improve over the medium-to-long term**, with the domestic primary deficit expected to more than double from 2.0 percent of GDP in 2014 to 4.5 percent in 2016, and then improve to 3.2 percent in 2019, stabilizing around 1.8 percent during 2020–34.

   - **The current account deficit is forecast to widen from 7.4 percent of GDP in 2014 to 13.0 percent in 2015 and then narrow over the medium-to-long term**, consistent with the projected production profile of iron ore and other exports. The projected import dynamics broadly reflect the expected overall real GDP growth.

   - **External debt is projected to increase from 28.2 percent of GDP in 2014 to 33.7 percent in 2015**, reflecting the augmentation of access to IMF financing. External debt will stabilize around 20 percent of GDP in the long run. The DSA also assumes some nonconcessional borrowing limited at 0.5 percent of GDP (US$30 million) per year during

the projection period to finance key priority projects for which the authorities would not be able to secure concessional funding.

- **Domestic debt is projected to rise from 9.2 percent of GDP in 2013, to around 20 percent by the end of the projection period**, mainly reflecting increased domestic borrowing to finance public investment.

3. **Compared with the previous DSA, the key changes in this updated DSA are as follows** (Text Table 1): (i) real GDP growth is expected to be lower in the short term, due to the effects of the Ebola epidemic and severe disruptions in the iron-ore sector and low global iron ore prices. Growth is projected to slow from 8 percent to 6 percent in 2014 and from 10.4 percent to -12.8 percent in 2015 (ii) a higher primary deficit, particularly in the short run, driven by Ebola-related revenue shortfalls and expanded spending needs; (iii) a higher level of domestic borrowing, in line with a weaker fiscal position; (iv) the external current account deficit is expected to be higher as export growth decelerates, in line with lower domestic production and declines in iron-ore prices; and (v) the exchange rate depreciates significantly in the short run, pushing up the ratio of external debt to revenue and GDP.

Text Table 1. Selected Economic Indicators, 2013-2034
(In percent of GDP, unless otherwise indicated)[1]

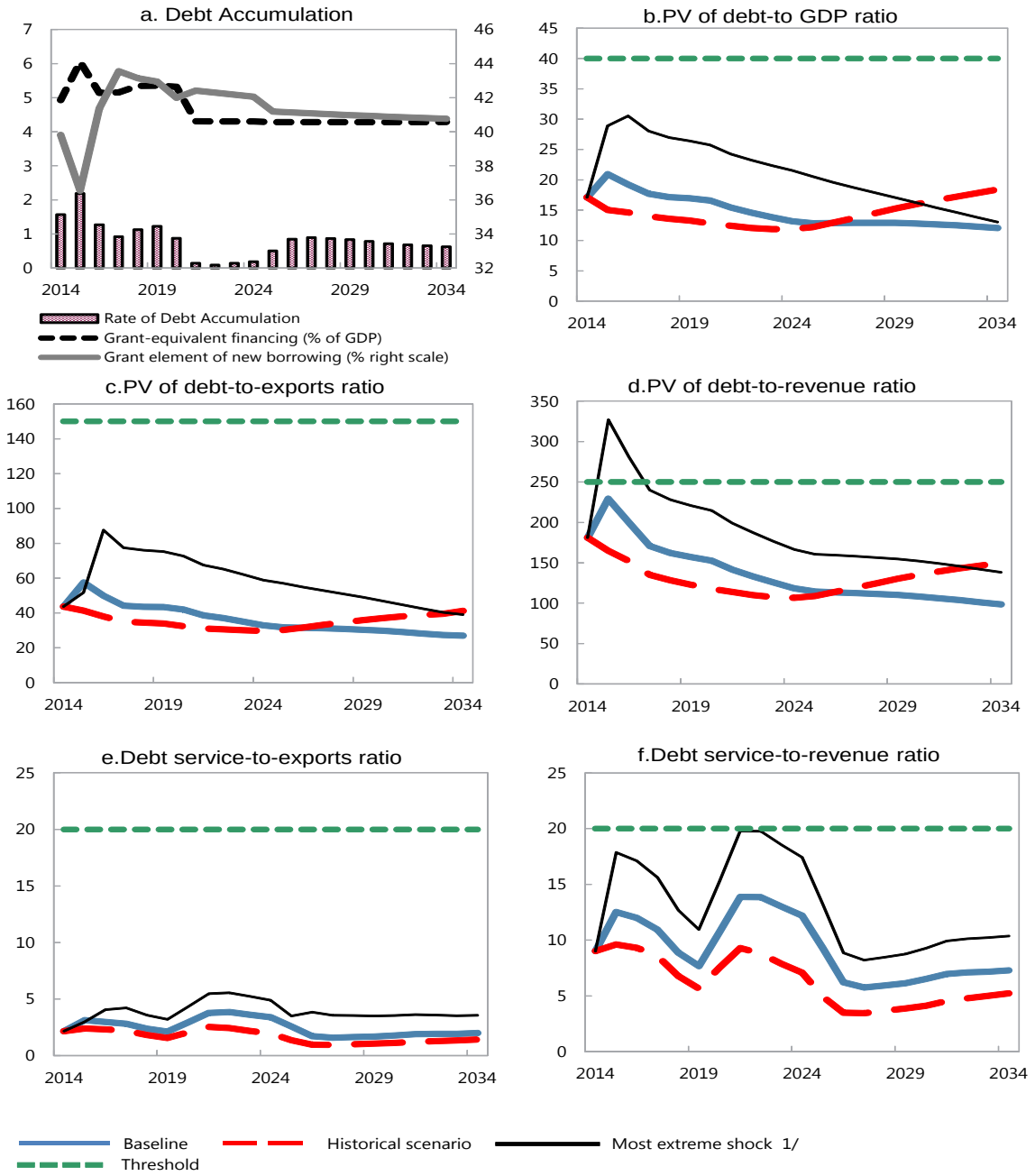
	2013	2014	2015	2016	2017	2018	2019	Long Term[2]
Real GDP Growth (in percent)								
Current DSA	20.1	6.0	-12.8	8.4	8.9	6.0	6.0	5.4
Previous DSA	20.1	8.0	10.4	7.8	4.8	4.8	4.9	5.4
Primary fiscal deficit								
Current DSA	1.1	2.0	3.5	4.5	3.7	3.3	3.2	1.8
Previous DSA	1.1	3.9	4.0	2.6	2.6	2.4	2.3	2.0
Current account deficit								
Current DSA	10.4	7.4	13.0	8.3	7.1	6.8	6.1	4.3
Previous DSA	10.4	10.8	7.9	6.5	6.7	6.3	5.9	4.2
Foreign direct investment								
Current DSA	7.3	4.6	3.0	5.6	4.5	4.0	3.3	3.2
Previous DSA	7.3	5.6	5.4	4.4	3.8	3.5	2.9	2.5
External debt								
Current DSA	25.2	28.2	33.7	30.5	27.7	26.5	25.9	19.9
Previous DSA	21.2	23.1	23.0	22.9	23.7	24.5	25.3	28.7
Domestic debt								
Current DSA	9.2	10.7	12.6	12.5	12.6	13.0	13.6	19.5
Previous DSA	9.2	9.5	9.6	9.8	10.2	10.4	10.3	11.3

Sources: The Sierra Leone Authorities and IMF staff projections.

1/ GDP includes iron ore activity.

2/ For the current DSA, the long term covers the period 2020-34, and for the previous DSA it covers the period 2020-34.

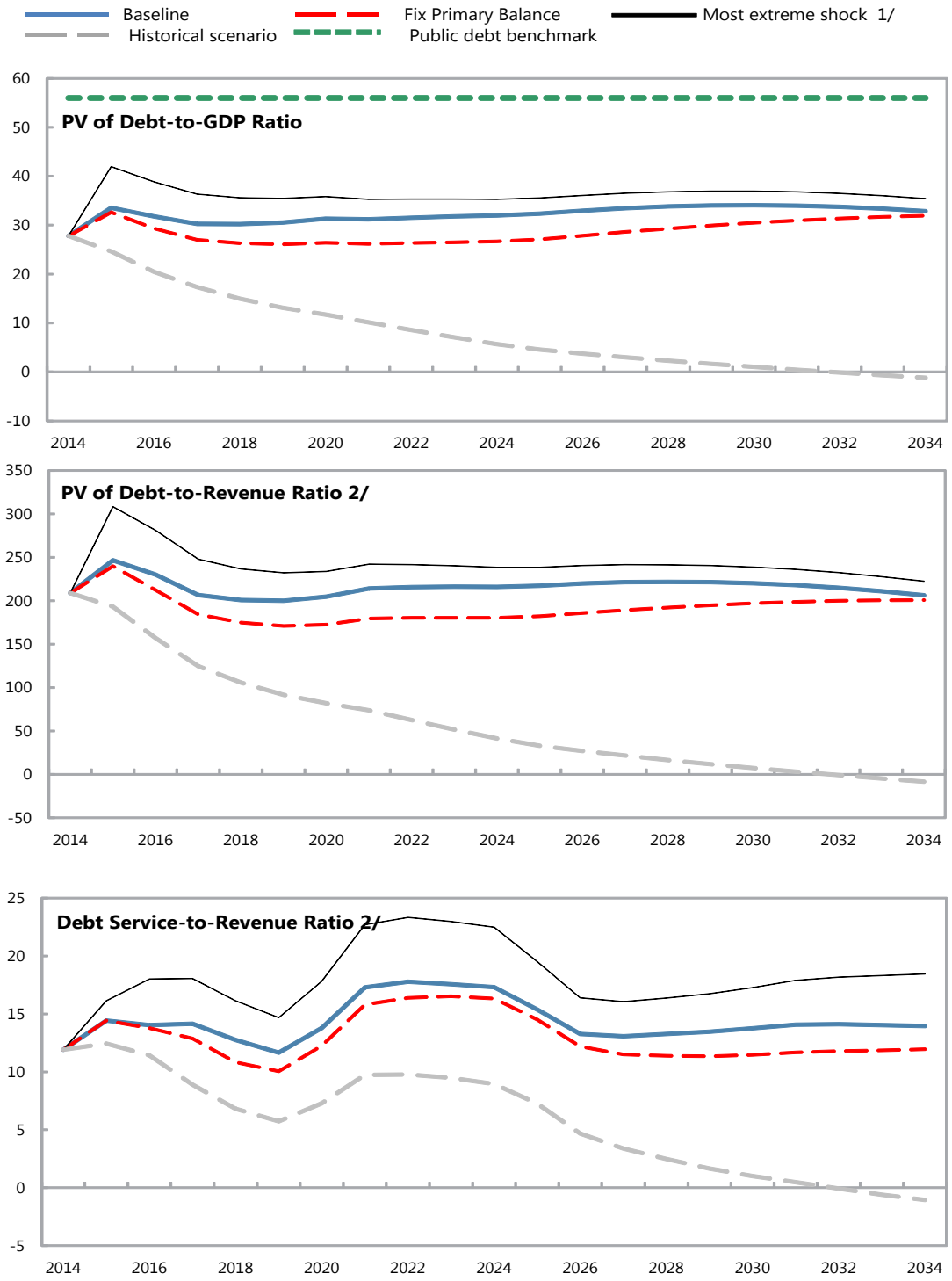
Figure 1. Sierra Leone: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Non-debt flows shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Sierra Leone: Indicators of Debt under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2011–34[1]

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**33.3**	**26.2**	**25.2**			**28.2**	**33.7**	**30.5**	**27.7**	**26.5**	**25.9**		**20.1**	**17.4**	
of which: public and publicly guaranteed (PPG)	33.3	26.2	25.2			28.2	33.7	30.5	27.7	26.5	25.9		20.1	17.4	
Change in external debt	1.3	-7.1	-1.0			3.0	5.5	-3.2	-2.8	-1.2	-0.7		-0.9	-0.3	
Identified net debt-creating flows	4.4	4.6	-2.9			1.5	9.8	0.9	0.5	1.5	1.6		0.8	-0.5	
Non-interest current account deficit	**44.7**	**28.6**	**10.4**	**13.2**	**13.9**	**7.4**	**8.4**	**8.4**	**7.1**	**6.7**	**6.2**		**4.9**	**3.0**	4.3
Deficit in balance of goods and services	52.1	30.2	3.2			4.9	11.7	7.6	2.5	0.9	0.3		-3.7	-10.3	
Exports	18.6	35.5	44.0			39.2	36.4	38.6	40.0	39.4	39.2		40.0	44.6	
Imports	70.7	65.8	47.2			44.1	48.0	46.2	42.6	40.3	39.4		36.3	34.3	
Net current transfers (negative = inflow)	-8.5	-6.2	-4.1	-6.4	1.8	-8.4	-14.0	-5.8	-5.7	-5.7	-5.7		-5.7	-6.1	-5.8
of which: official	-3.8	-2.5	-1.0			-3.1	-10.6	-1.8	-1.8	-1.8	-1.9		-2.0	-2.6	
Other current account flows (negative = net inflow)	1.0	4.6	11.4			11.0	10.8	6.6	10.3	11.5	11.6		14.3	19.4	
Net FDI (negative = inflow)	**-36.6**	**-16.7**	**-7.4**	**-9.7**	**10.6**	**-4.6**	**-3.0**	**-5.3**	**-4.3**	**-3.8**	**-3.2**		**-3.3**	**-2.7**	-3.1
Endogenous debt dynamics 2/	**-3.7**	**-7.4**	**-5.9**			**-1.3**	**4.3**	**-2.2**	**-2.2**	**-1.4**	**-1.3**		**-0.8**	**-0.7**	
Contribution from nominal interest rate	0.2	0.2	0.0			0.1	0.2	0.2	0.1	0.1	0.1		0.2	0.2	
Contribution from real GDP growth	-1.7	-3.9	-4.1			-1.5	4.1	-2.4	-2.4	-1.5	-1.5		-1.0	-0.9	
Contribution from price and exchange rate changes	-2.2	-3.6	-1.9			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.1**	**-11.6**	**1.8**			**1.5**	**-4.3**	**-4.1**	**-3.3**	**-2.8**	**-2.3**		**-1.7**	**0.1**	
of which: exceptional financing	0.0	0.0	0.0			0.0	-0.7	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	15.0			17.1	20.9	19.2	17.7	17.2	17.0		13.2	12.1	
In percent of exports	...	...	34.1			43.7	57.5	49.9	44.2	43.5	43.4		33.0	27.0	
PV of PPG external debt	**...**	**...**	**15.0**			**17.1**	**20.9**	**19.2**	**17.7**	**17.2**	**17.0**		**13.2**	**12.1**	
In percent of exports	**...**	**...**	**34.1**			**43.7**	**57.5**	**49.9**	**44.2**	**43.5**	**43.4**		**33.0**	**27.0**	
In percent of government revenues	**...**	**...**	**140.1**			**181.1**	**229.4**	**199.8**	**170.8**	**162.2**	**157.1**		**118.4**	**98.5**	
Debt service-to-exports ratio (in percent)	**3.3**	**1.6**	**0.0**			**2.2**	**3.1**	**3.0**	**2.8**	**2.4**	**2.1**		**3.4**	**2.0**	
PPG debt service-to-exports ratio (in percent)	**3.3**	**1.6**	**0.0**			**2.2**	**3.1**	**3.0**	**2.8**	**2.4**	**2.1**		**3.4**	**2.0**	
PPG debt service-to-revenue ratio (in percent)	**5.3**	**5.0**	**0.0**			**9.0**	**12.5**	**12.0**	**10.9**	**8.9**	**7.7**		**12.2**	**7.3**	
Total gross financing need (Billions of U.S. dollars)	0.3	0.5	0.2			0.2	0.3	0.2	0.2	0.2	0.3		0.3	0.2	
Non-interest current account deficit that stabilizes debt ratio	43.4	35.7	11.5			4.5	3.0	11.6	9.8	7.9	6.8		5.8	3.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.0	15.2	20.1	7.9	5.5	6.0	-12.8	8.4	8.9	6.0	6.0	3.8	5.0	5.6	5.4
GDP deflator in US dollar terms (change in percent)	7.3	12.2	7.6	5.5	6.0	-3.1	0.2	7.9	4.7	3.5	2.3	2.6	1.3	1.4	1.4
Effective interest rate (percent) 5/	0.7	0.7	0.0	0.8	0.5	0.6	0.6	0.7	0.5	0.5	0.5	0.6	0.9	1.2	1.0
Growth of exports of G&S (US dollar terms, in percent)	28.9	146.8	60.1	30.3	45.0	-8.5	-18.9	24.1	18.4	8.0	7.7	5.1	7.6	6.2	7.8
Growth of imports of G&S (US dollar terms, in percent)	90.1	20.1	-7.2	23.4	37.3	-4.0	-4.8	12.4	5.1	3.9	6.0	3.1	5.0	6.5	5.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	39.8	36.6	41.4	43.6	43.1	42.9	41.2	42.1	40.8	41.4
Government revenues (excluding grants, in percent of GDP)	11.5	11.4	10.7			9.5	9.1	9.6	10.4	10.6	10.8		11.1	12.3	11.5
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.2			0.3	0.3	0.3	0.3	0.4	0.4		0.5	0.9	
of which: Grants	0.2	0.1	0.1			0.2	0.2	0.2	0.3	0.3	0.3		0.4	0.7	
of which: Concessional loans	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.9	6.1	5.1	5.2	5.3	5.4		4.3	4.3	4.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			75.1	69.0	79.3	82.3	82.0	82.0		82.9	82.5	82.6
Memorandum items:															
Nominal GDP (Billions of US dollars)	2.9	3.8	4.9			5.0	4.4	5.2	5.9	6.4	7.0		9.8	18.9	
Nominal dollar GDP growth	13.7	29.2	29.3			2.7	-12.6	17.0	14.0	9.7	8.4	6.5	6.4	7.1	6.9
PV of PPG external debt (in Billions of US dollars)			0.7			0.8	0.9	1.0	1.0	1.1	1.2		1.3	2.3	
(PVt-PVt-1)/GDPt-1 (in percent)						1.6	2.2	1.3	0.9	1.1	1.2	1.4	0.2	0.6	0.6
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.1	0.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	14.9			17.0	20.7	19.1	17.6	17.0	16.9		13.1	12.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	33.5			42.8	56.0	48.8	43.3	42.8	42.6		32.5	26.8	
Debt service of PPG external debt (in percent of exports + remittance	...	...	0.0			2.1	3.1	2.9	2.8	2.3	2.1		3.4	2.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34[1]

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	17	21	19	18	17	17	**13**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	17	15	15	14	14	13	**12**	18
A2. New public sector loans on less favorable terms in 2014-2034 2	17	22	20	19	19	19	**17**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	17	18	17	16	15	15	**12**	11
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	17	20	24	22	22	21	**17**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	17	21	21	19	18	18	**14**	13
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	17	29	31	28	27	26	**22**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	17	13	10	9	9	9	**7**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	17	30	27	25	24	24	**19**	17
PV of debt-to-exports ratio								
Baseline	44	58	50	44	44	43	**33**	27
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	44	41	38	35	34	34	**30**	41
A2. New public sector loans on less favorable terms in 2014-2034 2	44	60	52	47	48	49	**43**	40
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	44	57	49	44	43	43	**32**	27
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	44	52	88	77	76	75	**59**	39
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	44	57	49	44	43	43	**32**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	44	79	79	70	68	67	**54**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	44	34	26	23	23	23	**16**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	44	57	49	44	43	43	**32**	27
PV of debt-to-revenue ratio								
Baseline	181	229	200	171	162	157	**118**	98
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	181	165	152	135	128	123	**107**	150
A2. New public sector loans on less favorable terms in 2014-2034 2	181	238	209	184	179	177	**153**	146
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	181	195	178	152	144	140	**105**	87
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	181	217	254	216	205	197	**153**	103
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	181	231	215	184	174	169	**127**	106
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	181	317	317	271	255	244	**193**	107
B5. Combination of B1-B4 using one-half standard deviation shocks	181	147	105	90	86	85	**59**	80
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	181	327	281	240	228	221	**166**	138

Table 2. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34[1] (Concluded)

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
Debt service-to-exports ratio								
Baseline	2	3	3	3	2	2	**3**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	2	2	2	2	2	2	**2**	1
A2. New public sector loans on less favorable terms in 2014-2034 2	2	3	3	3	3	2	**4**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	2	3	3	3	2	2	**3**	2
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	2	3	4	4	4	3	**5**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	2	3	3	3	2	2	**3**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	2	3	3	3	3	3	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	2	2	2	**3**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	2	3	3	3	2	2	**3**	2
Debt service-to-revenue ratio								
Baseline	9	13	12	11	9	8	**12**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	9	10	9	9	7	6	**7**	5
A2. New public sector loans on less favorable terms in 2014-2034 2	9	13	12	12	10	9	**13**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	9	11	11	10	8	7	**11**	7
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	9	13	12	12	10	8	**13**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	9	13	13	12	10	8	**13**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	9	13	13	13	11	9	**13**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	9	10	10	9	7	6	**10**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	9	18	17	16	13	11	**17**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	**40**	40

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	44.9	36.9	34.4			38.8	46.3	43.0	40.3	39.6	39.4		38.9	38.2	
of which: foreign-currency denominated	33.5	26.3	25.2			28.2	33.7	30.5	27.7	26.5	25.9		20.1	17.4	
Change in public sector debt	-1.9	-8.0	-2.5			4.4	7.5	-3.2	-2.7	-0.8	-0.1		-0.1	-0.6	
Identified debt-creating flows	-4.2	-6.0	-5.1			2.5	6.4	-1.2	-0.8	0.5	0.8		0.7	-0.6	
Primary deficit	2.6	3.5	1.1	-1.3	7.5	2.0	3.5	4.5	3.7	3.3	3.2	3.4	2.2	1.0	1.8
Revenue and grants	17.0	15.2	13.3			13.3	13.6	13.8	14.7	15.0	15.3		14.8	15.9	
of which: grants	5.6	3.8	2.6			3.8	4.5	4.2	4.3	4.5	4.5		3.7	3.7	
Primary (noninterest) expenditure	19.6	18.7	14.4			15.3	17.1	18.3	18.3	18.4	18.4		17.0	16.9	
Automatic debt dynamics	-6.1	-8.7	-6.1			0.9	3.2	-5.5	-4.5	-2.8	-2.3		-1.5	-1.6	
Contribution from interest rate/growth differential	-3.3	-5.8	-4.9			-1.8	4.7	-4.2	-3.9	-2.5	-2.3		-1.2	-1.4	
of which: contribution from average real interest rate	-0.7	0.1	1.2			0.2	-1.0	-0.6	-0.4	-0.2	0.0		0.7	0.7	
of which: contribution from real GDP growth	-2.6	-5.9	-6.2			-2.0	5.7	-3.6	-3.5	-2.3	-2.2		-1.9	-2.1	
Contribution from real exchange rate depreciation	-2.8	-2.9	-1.2			2.7	-1.5	-1.4	-0.6	-0.4	0.0		...	...	
Other identified debt-creating flows	-0.7	-0.8	-0.1			-0.3	-0.3	-0.1	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.3	-0.1	-0.1			-0.1	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.5	-0.8	0.0			-0.2	-0.2	-0.1	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.3	-2.0	2.6			1.9	1.0	-2.1	-1.9	-1.3	-1.0		-0.8	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	24.2			27.8	33.6	31.8	30.3	30.2	30.5		32.0	32.8	
of which: foreign-currency denominated	...	...	15.0			17.1	20.9	19.2	17.7	17.2	17.0		13.2	12.1	
of which: external	...	...	15.0			17.1	20.9	19.2	17.7	17.2	17.0		13.2	12.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.0	5.6	2.4			3.5	5.5	6.4	5.8	5.3	4.9		4.7	3.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	182.3			209.0	246.7	230.0	206.5	200.7	200.0		216.1	206.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	226.1			294.0	367.8	330.3	292.3	285.3	282.6		287.0	268.1	
of which: external 3/	...	...	140.1			181.1	229.4	199.8	170.8	162.2	157.1		118.4	98.5	
Debt service-to-revenue and grants ratio (in percent) 4/	14.0	13.9	9.4			11.9	14.4	14.0	14.2	12.7	11.7		17.3	14.0	
Debt service-to-revenue ratio (in percent) 4/	20.8	18.6	11.7			16.8	21.5	20.1	20.0	18.1	16.5		23.0	18.1	
Primary deficit that stabilizes the debt-to-GDP ratio	4.5	11.5	3.6			-2.5	-4.0	7.7	6.4	4.1	3.3		2.3	1.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.0	15.2	20.1	7.9	5.5	6.0	-12.8	8.4	8.9	6.0	6.0	3.8	5.0	5.6	5.4
Average nominal interest rate on forex debt (in percent)	0.7	0.7	0.0	0.9	0.6	0.6	0.6	0.7	0.5	0.5	0.5	0.6	0.9	1.2	1.0
Average real interest rate on domestic debt (in percent)	-1.9	4.8	7.0	-0.7	4.2	4.8	-5.8	-2.1	0.9	2.0	2.7	0.4	2.8	2.4	2.7
Real exchange rate depreciation (in percent, + indicates depreciation	-9.3	-10.0	-5.2	-3.6	8.5	11.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	17.3	12.0	7.7	12.1	4.0	3.8	14.3	9.7	7.8	6.6	5.3	7.9	4.4	4.4	4.4
Growth of real primary spending (deflated by GDP deflator, in percer	11.2	9.8	-7.5	1.4	5.3	12.3	-2.2	16.1	9.1	6.2	6.3	8.0	5.0	5.7	4.9
Grant element of new external borrowing (in percent)	...	...	...	...	...	39.8	36.6	41.4	43.6	43.1	42.9	41.2	42.1	40.8	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Sierra Leone: Sensitivity Analysis for Key Indicators of Public Debt, 2014–34

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	28	34	32	30	30	31	32	33
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	25	20	17	15	13	6	-1
A2. Primary balance is unchanged from 2014	28	33	29	27	26	26	27	32
A3. Permanently lower GDP growth 1/	28	34	33	32	32	33	40	58
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	28	28	27	25	25	25	23	20
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	28	35	34	33	32	33	34	34
B3. Combination of B1-B2 using one half standard deviation shocks	28	28	26	24	23	23	20	14
B4. One-time 30 percent real depreciation in 2015	28	42	39	36	36	35	35	35
B5. 10 percent of GDP increase in other debt-creating flows in 2015	28	40	37	35	35	35	36	35
PV of Debt-to-Revenue Ratio 2/								
Baseline	209	247	230	206	201	200	216	206
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	209	193	157	125	106	92	41	-8
A2. Primary balance is unchanged from 2014	209	240	212	184	175	171	180	201
A3. Permanently lower GDP growth 1/	209	250	236	215	212	215	263	342
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	209	215	204	179	170	166	161	127
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	209	259	249	222	215	213	228	213
B3. Combination of B1-B2 using one half standard deviation shocks	209	215	197	171	160	154	139	90
B4. One-time 30 percent real depreciation in 2015	209	308	281	248	237	232	239	223
B5. 10 percent of GDP increase in other debt-creating flows in 2015	209	292	270	240	231	229	243	220
Debt Service-to-Revenue Ratio 2/								
Baseline	12	14	14	14	13	12	17	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	12	11	9	7	6	9	-1
A2. Primary balance is unchanged from 2014	12	14	14	13	11	10	16	12
A3. Permanently lower GDP growth 1/	12	15	14	15	13	12	20	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	12	13	13	12	11	10	15	9
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	12	14	14	16	14	12	18	15
B3. Combination of B1-B2 using one half standard deviation shocks	12	13	12	12	10	9	13	7
B4. One-time 30 percent real depreciation in 2015	12	16	18	18	16	15	22	18
B5. 10 percent of GDP increase in other debt-creating flows in 2015	12	14	16	20	13	13	18	16

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.